



Tube-Based MPC for Uncertain Sampled-Data Control Systems With Inter-Sample Reachability Analysis

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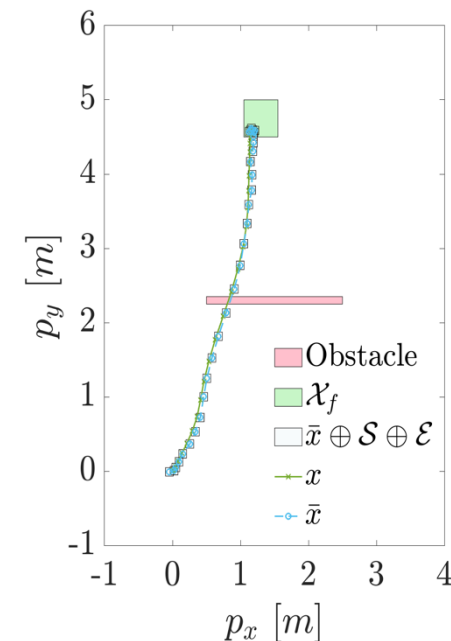
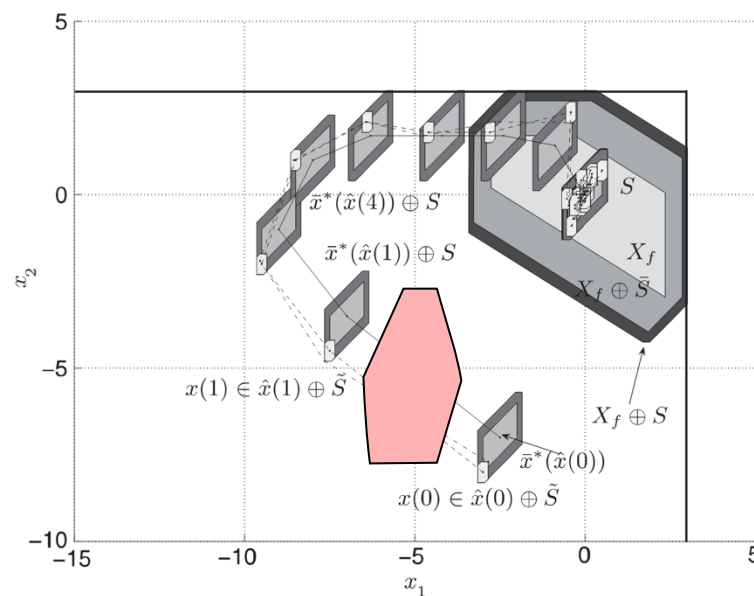
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Background

- MPC for **Constrained** Systems (with Unknown but **Bounded Uncertainties**)

Tube-based MPC

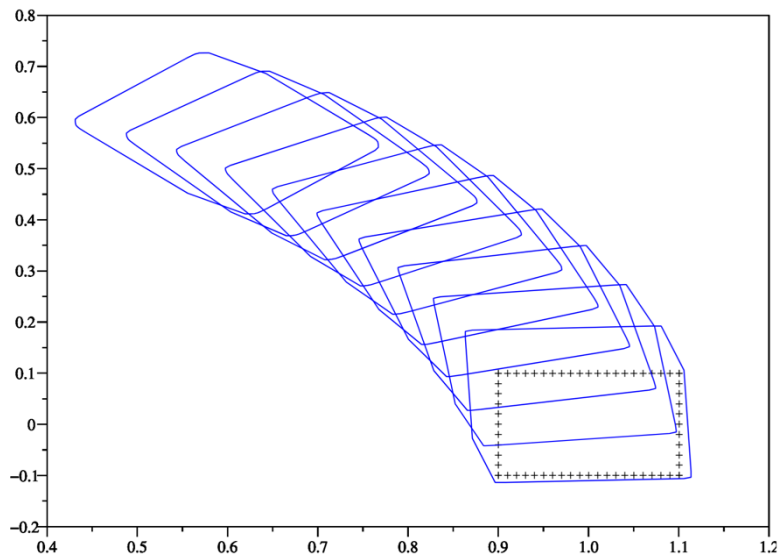


Guaranteed safety (at **sampled** time)

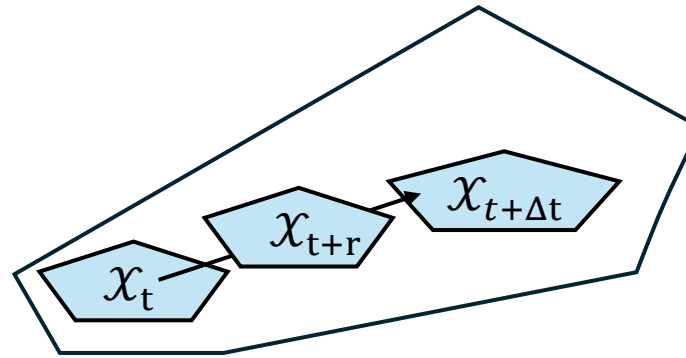
Unsafe for **inter-sample** trajectories

Background

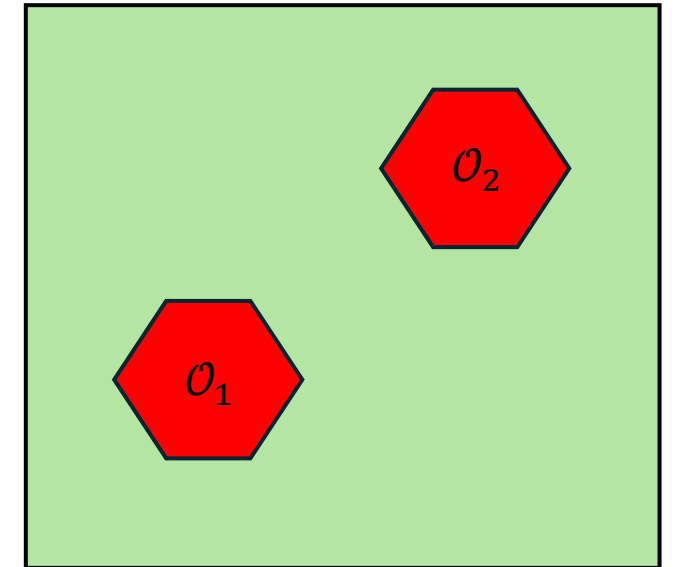
- Reachability Analysis



Zonotope (**over conservative**)



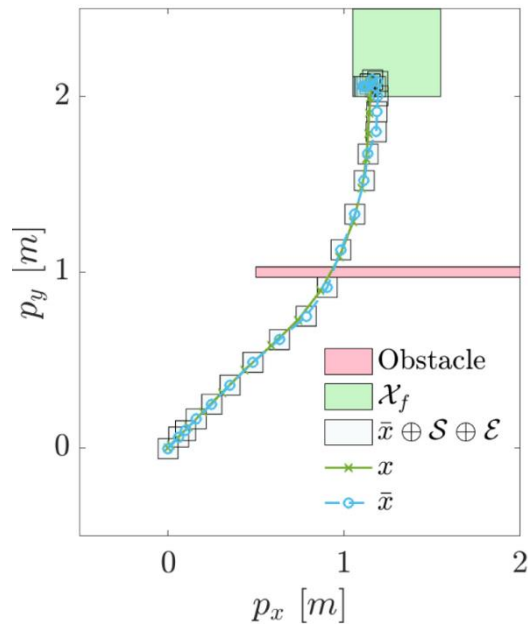
$$\mathcal{R}_{[t, t+\Delta t]} = \bigcup_{r \in [0, \Delta t]} x_{t+r}$$



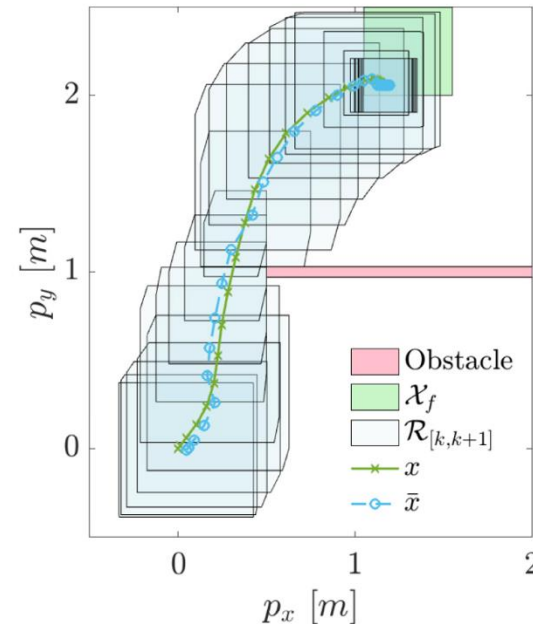
Non-convex environment
(**no explicit form**)

Background

- Goal:
 - Tube-MPC
 - Reachability analysis (polytope and a tighter bound)
 - Explicit form for non-convex constraints (obstacle avoidance)



Without Reachability



With Reachability

Problem 1 (Output-Feedback Robust MPC With Reachability Analysis)

$$\begin{aligned}
 & J = \min_{u_k, x_k} \sum_{k=0}^{N-1} \ell(x_k, u_k) + V_f(x_N) \\
 \text{s.t.} \quad & \text{Dynamics} \quad \left\{ \begin{array}{l} \dot{x}(t) = Ax(t) + Bu(t) + Ww(t) + f \\ z_k = Cx_k + Du_k + Vv_k + g \end{array} \right. \\
 & \text{State} \quad \left\{ \begin{array}{l} x_0 \in \mathcal{X}_0 \triangleq \{x \mid P_0 x \leq p_0\} \\ x \in \mathcal{X} \triangleq \{x \mid P_x x \leq p_x\} \\ x_N \in \mathcal{X}_f \triangleq \{x \mid P_f x \leq p_f\} \end{array} \right. \\
 & \text{Input} \quad \left\{ \begin{array}{l} u \in \mathcal{U} \triangleq \{u \mid P_u u \leq p_u\} \end{array} \right. \\
 & \text{Noise} \quad \left\{ \begin{array}{l} w \in \mathcal{W} \triangleq \{w \mid Q_w w \leq q_w\} \\ v \in \mathcal{V} \triangleq \{v \mid Q_v v \leq q_v\} \end{array} \right. \\
 & \text{Obs avoidance} \quad \left\{ \begin{array}{l} x \in \mathcal{X} \setminus \bigcup_{j=1}^{n_{\otimes}} \mathcal{O}_j \end{array} \right.
 \end{aligned}$$

Methods:

Time-discretization

Tube-based Method

- Constraint tightening
- Reachability analysis (polytope and a tighter bound)

Non-convex Environment

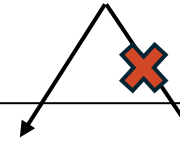
- Time-discretization

Continuous-time Dynamics:

$$\dot{x}(t) = Ax(t) + Bu(t) + Ww(t) + f$$

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}(Bu(\tau) + Ww(\tau) + f)d\tau$$

Zero-order hold



$$\tilde{w}_k = \max_{\tau \in [k\Delta t, (k+1)\Delta t]} |w(\tau)|$$

Discrete-time Dynamics:

$$x_{k+1}^a = A_d x_k^a + B_d u_k + W_d \tilde{w}_k + f_d$$

$$\begin{aligned} A_d &= e^{A\Delta t} \\ B_d &= \int_0^{\Delta t} e^{A\tau} d\tau B \\ W_d &= \int_0^{\Delta t} |e^{A\tau} W| d\tau \\ f_d &= \int_0^{\Delta t} e^{A\tau} d\tau f \end{aligned}$$

- Output-feedback Robust MPC

State-estimator:

$$\hat{x}_{k+1} = A_d \hat{x}_k + B_d u_k + f_d + L(z_k - C \hat{x}_k - D u_k - g)$$

L : observer gain $\rightarrow A_d - LC$ Schur stable

Nominal dynamics:

$$\bar{x}_{k+1} = A_d \bar{x}_k + B_d \bar{u}_k + f_d$$

without disturbance

Output feedback law:

$$u_k = \bar{u}_k - K(\hat{x}_k - \bar{x}_k)$$

K : controller gain $\rightarrow A_d - B_d K$ is Schur stable

- Constraint Tightening ($x, u \rightarrow \bar{x}, \bar{u}$)

- Estimation error:

$$\tilde{x}^a \triangleq x^a - \hat{x}$$

$$\tilde{x}_{k+1}^a = (A_d - LC)\tilde{x}_k^a + W_d \tilde{w}_k - LV v_k$$

- Robust Positively Invariant (RPI) set:

$$\tilde{x}_k^a \in \mathcal{E} = \bigoplus_{i=0}^{\infty} (A_d - LC)^i (W_d \mathcal{W} \oplus (-LV \mathcal{V}))$$

- Tracking error:

$$s \triangleq \hat{x} - \bar{x}$$

$$s_{k+1} = (A_d - B_d K)s_k + LC \tilde{x}_k + LV v_k$$

$$s_k \in \mathcal{S} = \bigoplus_{i=0}^{\infty} (A_d - B_d K)^i (LC \mathcal{E} \oplus LV \mathcal{V})$$

One-step optimization (to obtain $\mathcal{E}$ and $\mathcal{S}$)

Theorem 6. The zonotope $Z = \{\mathbf{G}\Phi, \mathbf{c}\} \subset \mathbb{R}^n$, with $\Phi = \text{diag}(\phi)$, $\phi_i > 0, \forall i \in \{1, \dots, n_g\}$, is an RPI set of (30) if $W = \{\mathbf{G}_w, \mathbf{c}_w\}$ and there exist $\Gamma_1 \in \mathbb{R}^{n_g \times n_g}$, $\Gamma_2 \in \mathbb{R}^{n_g \times n_w}$, and $\beta \in \mathbb{R}^{n_g}$ such that

$$\mathbf{A}\mathbf{G}\Phi = \mathbf{G}\Gamma_1, \quad (33a)$$

$$\mathbf{G}_w = \mathbf{G}\Gamma_2, \quad (33b)$$

$$(\mathbf{I} - \mathbf{A})\mathbf{c} - \mathbf{c}_w = \mathbf{G}\beta, \quad (33c)$$

$$|\Gamma_1| \mathbf{1} + |\Gamma_2| \mathbf{1} + |\beta| \leq \Phi \mathbf{1}. \quad (33d)$$

- Constraint Tightening ($\mathcal{X}, \mathcal{U} \rightarrow \bar{\mathcal{X}}, \bar{\mathcal{U}}$)

mRPI:

$$\tilde{x}_k \in \mathcal{E} = \{\tilde{x} \in \mathbb{R}^n : P_{\tilde{x}} \tilde{x} \leq p_{\tilde{x}}\}$$

$$s_k \in \mathcal{S} = \{s \in \mathbb{R}^n : P_s s \leq p_s\}$$

State:

$$x_k = \hat{x}_k + \tilde{x}_k \in \{\hat{x}_k\} \oplus \mathcal{E}$$

$$x_k = \bar{x}_k + s_k + \tilde{x}_k \in \{\bar{x}_k\} \oplus (\mathcal{S} \oplus \mathcal{E})$$

Control:

$$u_k = \bar{u}_k - K(\hat{x}_k - \bar{x}_k)$$

Set Operation:

$$\begin{aligned} \mathcal{U} \oplus \mathcal{V} &\triangleq \{u + v \mid u \in \mathcal{U}, v \in \mathcal{V}\} \\ \mathcal{U} \ominus \mathcal{V} &\triangleq \{u \mid u \oplus \mathcal{V} \subseteq \mathcal{U}\} \end{aligned}$$

Nominal state:

$$\bar{x}_0 \in \bar{\mathcal{X}}_0 = \{\hat{x}_0\} \oplus (-\mathcal{S}) \triangleq \{\bar{x} \in \mathbb{R}^n \mid Q_0 \bar{x} \leq q_0\}$$

$$\bar{x}_k \in \bar{\mathcal{X}} = \mathcal{X} \ominus (\mathcal{S} \oplus \mathcal{E}) \triangleq \{\bar{x} \in \mathbb{R}^n \mid Q_x \bar{x} \leq q_x\}$$

$$\bar{x}_N \in \bar{\mathcal{X}}_f = \mathcal{X}_f \ominus (\mathcal{S} \oplus \mathcal{E}) \triangleq \{\bar{x} \in \mathbb{R}^n \mid Q_f \bar{x} \leq q_f\}$$

Nominal control:

$$\bar{u}_k \in \bar{\mathcal{U}} = \mathcal{U} \ominus (-KS) \triangleq \{\bar{u} \in \mathbb{R}^{m_u} \mid Q_u \bar{u} \leq q_u\}$$

Lemma 1 (Approximation of $\mathcal{X}_{t+r}$)

Given the state set $\mathcal{X}_t$ at time t and step k , $\forall r \in [0, \Delta t]$, the state reachable set $\mathcal{X}_{t+r} = \Phi_r(\mathcal{X}_t)$ at $t + r$ can be over-approximated by:

$$\mathcal{X}_{t+r} \subseteq \boxed{e^{Ar} \mathcal{X}_t} \oplus \boxed{\square_p(\beta(r) + \gamma(r))}, \text{ variant with } r$$

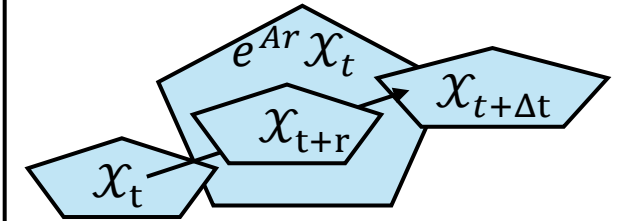
where

$$\beta(r) = \Lambda(r) \|B\|_p \|\bar{u}_k\|_p,$$

$$\gamma(r) = \Lambda(r) \bar{d}_p,$$

$$\Lambda(t) = \begin{cases} t, & \text{if } \mu_p(A) = 0 \\ \frac{e^{\mu_p(A)t} - 1}{\mu_p(A)}, & \text{otherwise,} \end{cases}$$

$$\bar{d}_p \triangleq \sup_{s \in \mathcal{S}, w \in \mathcal{W}} |(-BKs) + Ww + f|_p$$



$$\mathcal{R}_{[t, t+\Delta t]} = \bigcup_{r \in [0, \Delta t]} \mathcal{X}_{t+r}$$

$\|x\|_p$: vector norm with $p \in \{1, \infty\}$

$\square_p(r)$: hypercube with radius r

Lemma 2 (Approximation of $e^{Ar} \mathcal{X}_t$)

Given a set $\mathcal{X}_t$ at time t , $\forall r \in [0, \Delta t]$, $e^{Ar} \mathcal{X}_t$ can be over-approximated by:

$$e^{Ar} \mathcal{X}_t \subseteq \mathcal{P}_t \oplus \square_p(\alpha_{\Delta t}),$$

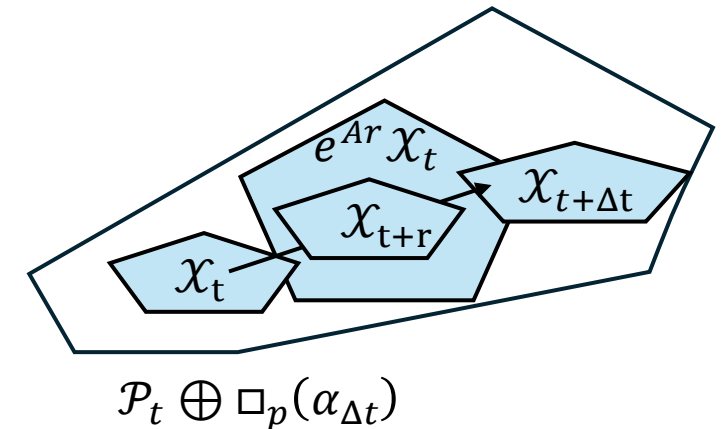
where

$$\mathcal{P}_t = \text{CH}\{\mathcal{X}_t, e^{A\Delta t} \mathcal{X}_t\},$$

$$\alpha_{\Delta t} = \kappa \sup_{x(t) \in \mathcal{X}_t} \|x(t)\|_p,$$

$$\kappa = \min\{\theta, e^{|A|_p \Delta t} - 1 - |A|_p \Delta t\},$$

$$\theta = \begin{cases} \frac{\|A\|_p^2 \Delta t^2}{2\sqrt{3}}, & \text{if } \mu_p(A) = 0 \\ \frac{1}{2} \|A\|_p^{2\sqrt{\Delta t}} \left(\frac{\mu_p(A)(\Lambda(\Delta t))^2 - 2\Lambda(\Delta t) + 2\Delta t}{2\mu_p(A)^2} \right)^{1/2}, & \text{otherwise.} \end{cases}$$



θ is tighter, using **logarithmic norm** and **Cauchy-Schwarz inequality**

Lemma 1 (Approximation of $\mathcal{X}_{t+r}$)

Given the state set $\mathcal{X}_t$ at time t and step k , $\forall r \in [0, \Delta t]$, the state reachable set $\mathcal{X}_{t+r} = \Phi_r(\mathcal{X}_t)$ at $t + r$ can be over-approximated by:

$$\mathcal{X}_{t+r} \subseteq e^{Ar} \mathcal{X}_t \oplus \square_p(\beta(r) + \gamma(r))$$

Lemma 2 (Approximation of $e^{Ar} \mathcal{X}_t$)

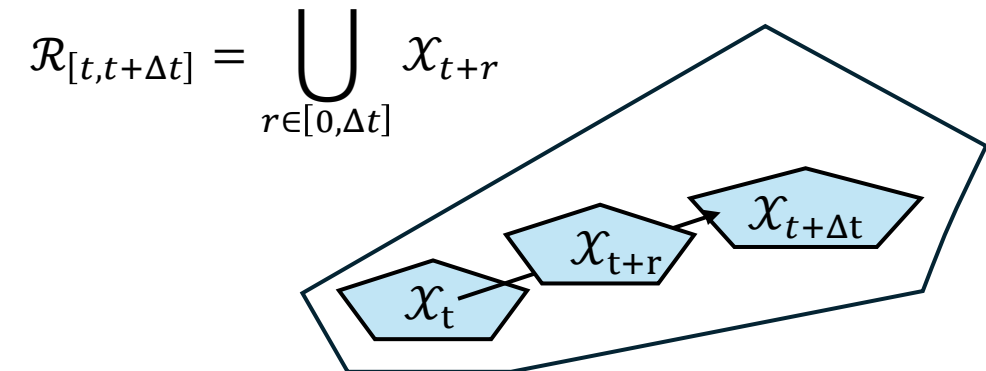
Given a set $\mathcal{X}_t$ at time t , $\forall r \in [0, \Delta t]$, $e^{Ar} \mathcal{X}_t$ can be over-approximated by:

$$e^{Ar} \mathcal{X}_t \subseteq \mathcal{P}_t \oplus \square_p(\alpha_{\Delta t})$$

Proposition 1 (Inter-Sample Reachable Set)

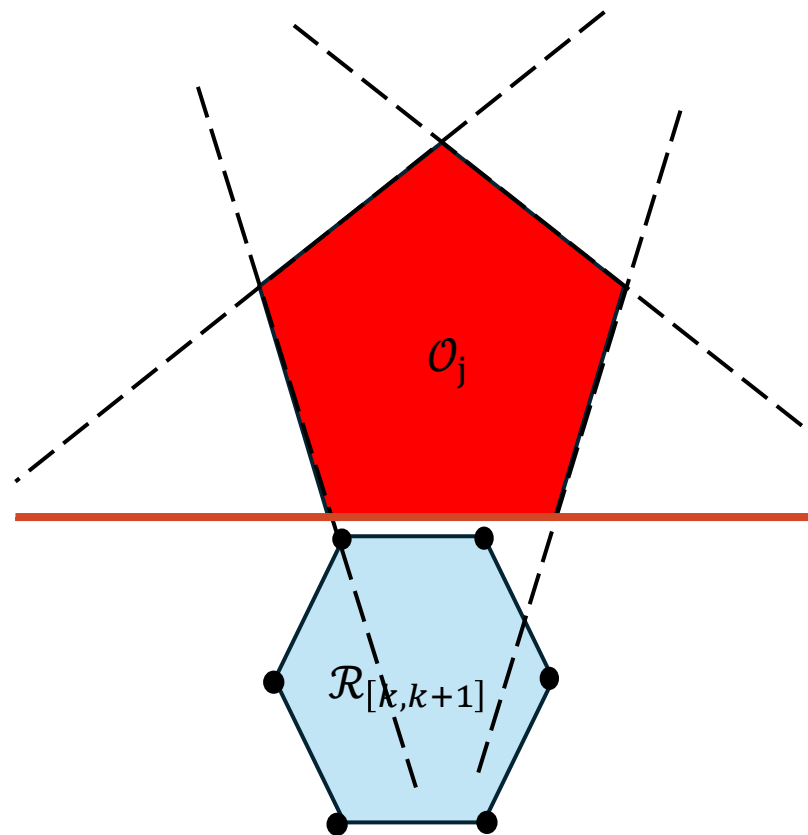
Given the continuous-time dynamics, state set $\mathcal{X}_t$, nominal state $\bar{x}(t) = \bar{x}_k$, and nominal input $\bar{u}(t) = \bar{u}_k$ at time t and step k , as well as the process noise $\mathcal{W}$ and tracking error $\mathcal{S}$, the reachable set $\mathcal{R}_{[t, t+\Delta t]}(\mathcal{X}_t)$ on the time interval $[t, t + \Delta t]$ can be over-approximated as follows:

$$\mathcal{R}_{[t, t+\Delta t]}(\mathcal{X}_t) \subseteq \mathcal{P}_t \oplus \square_p(\alpha_{\Delta t} + \beta(\Delta t) + \gamma(\Delta t))$$



Method

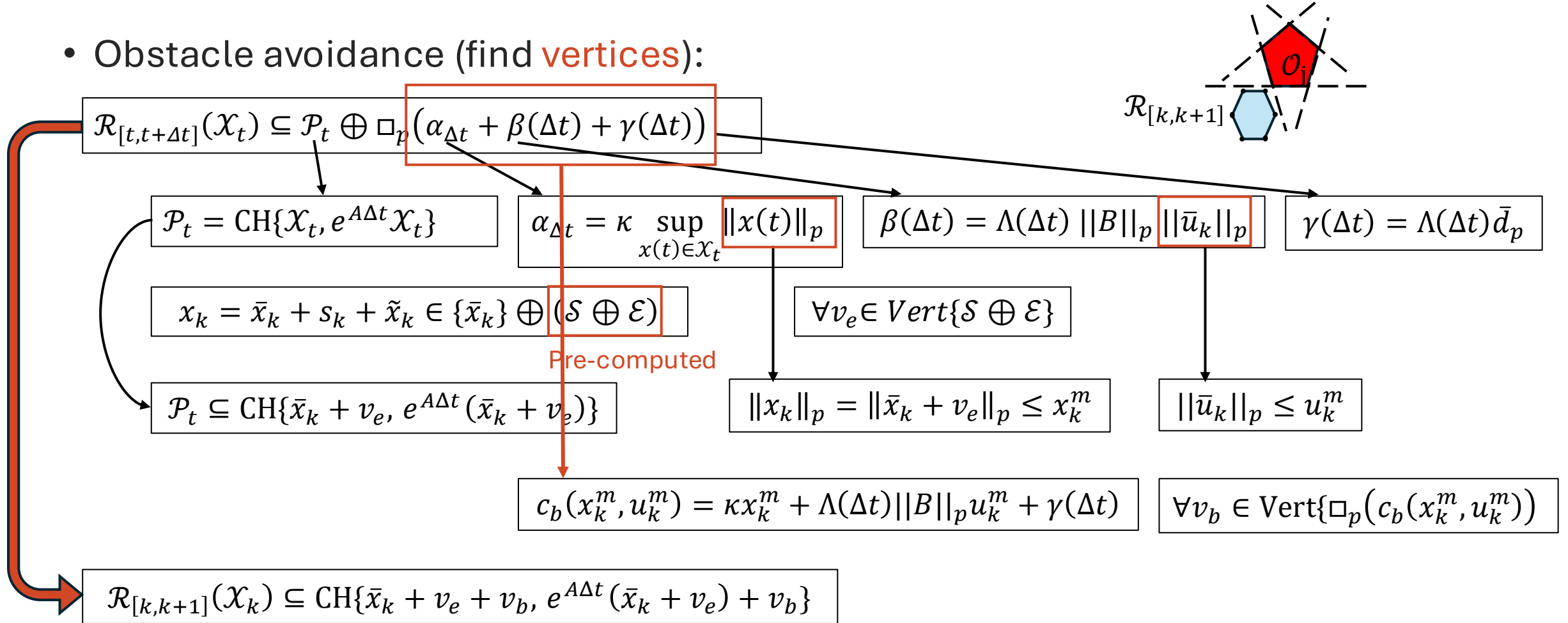
- Obstacle avoidance:



$$\mathcal{O}_j \triangleq \{x \mid P_{\otimes}^j x \leq p_{\otimes}^j\}, j \in \mathbb{Z}_{n_{\otimes}}^+$$

Keep all **vertices** outside of obstacles

- Obstacle avoidance (find **vertices**):



- Obstacle avoidance:

$$\mathcal{R}_{[k,k+1]}(\mathcal{X}_k) \subseteq \text{CH}\{\bar{x}_k + v_e + v_b, e^{A\Delta t}(\bar{x}_k + v_e) + v_b\}$$

δ_k : binary variable (active **constraint selector**)

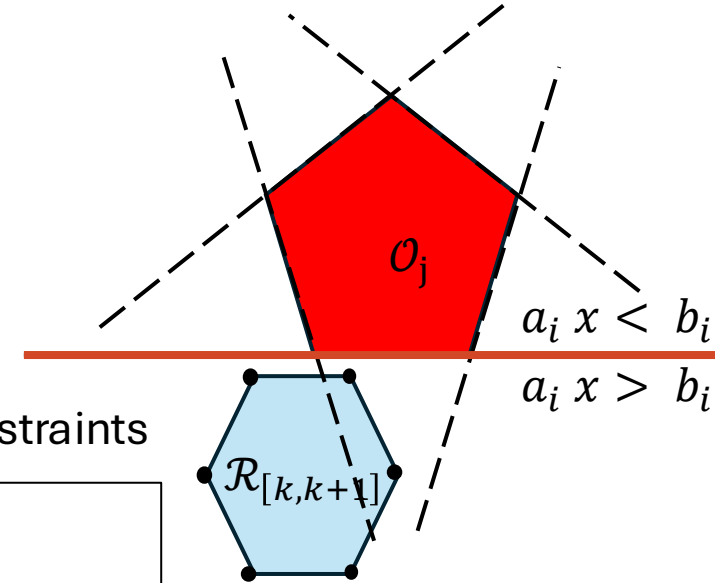
$$\begin{aligned} \delta_k^\top \mathbf{1} &\geq 1, \\ P_{\otimes}^j(\bar{x}_k + v_e + v_b) &\geq p_{\otimes}^j, \\ P_{\otimes}^j(e^{A\Delta t}(\bar{x}_k + v_e) + v_b) &\geq p_{\otimes}^j \end{aligned}$$

Vertices outside of **active** constraints

η_k^j : **slack variable** to relax constraints

$$\begin{aligned} \delta_k^\top \mathbf{1} &\geq 1, \\ \text{SOS-1: } \{\delta_k^j, \eta_k^j\}, \\ P_{\otimes}^j(\bar{x}_k + v_e + v_b) + \eta_k^j &\geq p_{\otimes}^j, \\ P_{\otimes}^j(e^{A\Delta t}(\bar{x}_k + v_e) + v_b) + \eta_k^j &\geq p_{\otimes}^j \end{aligned}$$

Vertices can violate **inactive** constraints



Theorem 1 (Continuous-Time Tube-based MPC)

A feasible and computationally tractable solution to Problem 1 can be obtained by finding the optimal $\bar{u}_k$ at each time step as the solution to the following optimization problem:

$$\begin{aligned}
 & \min_{\bar{x}_k, \bar{x}_r, x_k^m, u_k^m, \bar{u}_k, \delta_k, \eta_k} J(\bar{x}, \bar{u}, \bar{x}_r) = \sum_{k=0}^{N-1} \ell(\bar{x}_k, \bar{u}_k, \bar{x}_r) + V_f(\bar{x}_N, \bar{x}_r) \\
 & \text{s.t.} \quad \left. \begin{aligned}
 & \text{Nominal dynamics} \quad \bar{x}_{k+1} = A_d \bar{x}_k + B_d \bar{u}_k + f_d, \\
 & \text{Nominal input} \quad Q_u \bar{u}_k \leq q_u, \|\bar{u}_k\|_p \leq u_k^m \\
 & \text{State} \quad \left. \begin{aligned}
 & Q_0 \bar{x}_0 \leq q_0, Q_f \bar{x}_r \leq q_f, \\
 & \|\bar{x}_k + v_e\|_p \leq x_k^m, \\
 & P_x(\bar{x}_k + v_e + v_b) \leq p_x, \\
 & P_x(e^{A\Delta t}(\bar{x}_k + v_e) + v_b) \leq p_x,
 \end{aligned} \right\} \quad \begin{aligned}
 & Q_f \bar{x}_N \leq q_f, \\
 & \forall k \in \mathbb{Z}_{N-1}^0, \\
 & \forall j \in \mathbb{Z}_{N\otimes}^+, \\
 & \forall v_e \in \text{Vert}\{\mathcal{S} \oplus \mathcal{E}\}, \\
 & \forall v_b \in \text{Vert}\{\square_p(c_b(x_k^m, u_k^m))\}
 \end{aligned} \\
 & \text{Obs avoidance} \quad \left. \begin{aligned}
 & \delta_k^\top \mathbf{1} \geq 1, \text{SOS} - 1: \{\delta_k^j, \eta_k^j\}, \\
 & P_\otimes^j(\bar{x}_k + v_e + v_b) + \eta_k^j \geq p_\otimes^j, \\
 & P_\otimes^j(e^{A\Delta t}(\bar{x}_k + v_e) + v_b) + \eta_k^j \geq p_\otimes^j
 \end{aligned} \right\}
 \end{aligned}
 \right\} \quad \forall k \in \mathbb{Z}_{N-1}^0
 \end{aligned}$$

- Dynamics (Double Integrator):

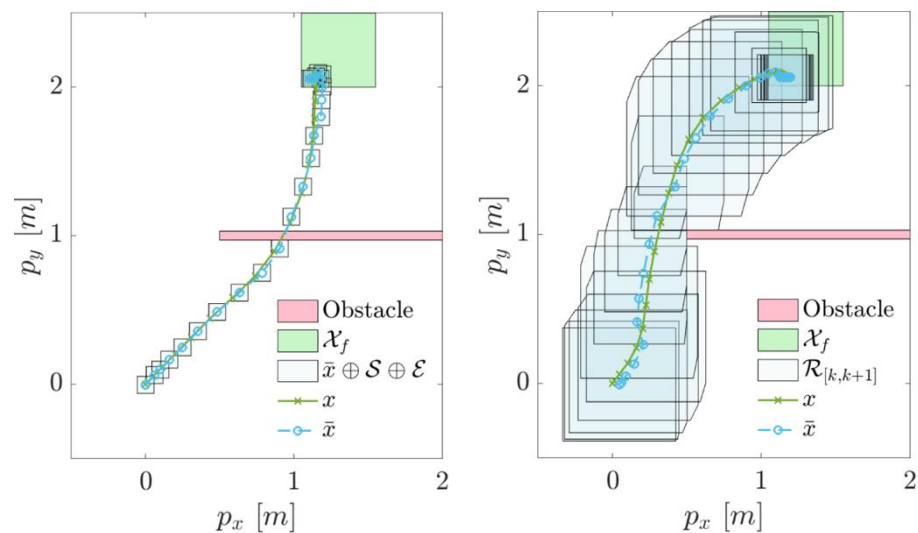
$$\dot{x}(t) = Ax(t) + Bu(t) + Ww(t)$$

$$z_k = Cx_k + Du_k + Vv_k$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{D_x}{M} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\frac{D_y}{M} \end{bmatrix} \quad B = W = \begin{bmatrix} 0 & 0 \\ \frac{1}{M} & 0 \\ 0 & 0 \\ 0 & \frac{1}{M} \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

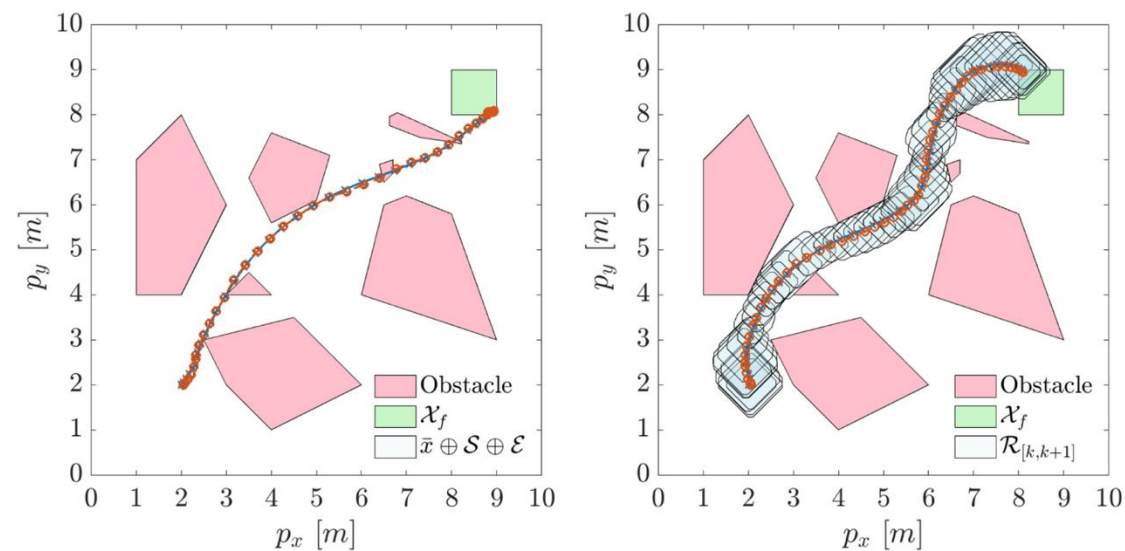
Single Obstacle



Without Reachability

With Reachability

Multiple Obstacles



Without Reachability

With Reachability

- We presented
 - A rigorous **time-discretization**
 - Reachability analysis with **polytopes** and **a tighter bound**
 - Tube-based MPC with inter-sample reachability analysis in **a non-convex environment**
- Future Work:
 - Extension to nonlinear systems
 - Trade-off between sample time and the size of reachable sets

References

- [1] S. Jafarpour, Z. Liu, and Y. Chen, “Probabilistic reachability analysis of stochastic control systems,” *IEEE Transactions on Automatic Control*, 2025.
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- [3] A. Girard, “Reachability of uncertain linear systems using zonotopes,” in *International workshop on hybrid systems: Computation and control*. Springer, 2005, pp. 291–305.
- [4] V. Raghuraman and J. P. Koeln, “Set operations and order reductions for constrained zonotopes,” *Automatica*, vol. 139, p. 110204, 2022.

Thank you!

Q & A

